SAR Lab 7: Landslide Time-Series Analysis

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Project Initialization

I started by setting up a Conda environment to install MintPy and its required packages. Next, I used the `download_licsar_pre_event.py` script to download the pre-processed COMET-LICS interferograms. I kept the default cutoff date (20170624) and used a maximum time gap of 72 days.

After getting the data, I created a `mintpy/` folder using the `licsar_to_mintpy_h5.py` script. I set the bounding box to `103.60 32.00 103.75 32.15`. This area perfectly covers the 2017 Xinmo landslide and the stable ground around it.

MintPy Processing

I ran the SBAS inversion using the `smallbaselineApp.py` script. For the analysis, I focused on the results saved in the `mintpy/` folder, especially the `velocity.h5` and `timeseries.h5` files.

At first, using the automatic reference point added a lot of noise and seasonal changes to the data. To fix this, I looked at the coherence and velocity maps to manually pick a new, stable reference point just outside the landslide area. Using this stable point in the settings gave much cleaner and more reliable results.

Displacement Time-Series Analysis

The main goal of this analysis was to measure the movement of the Xinmo slope before the event. The MintPy results clearly show the active landslide area.

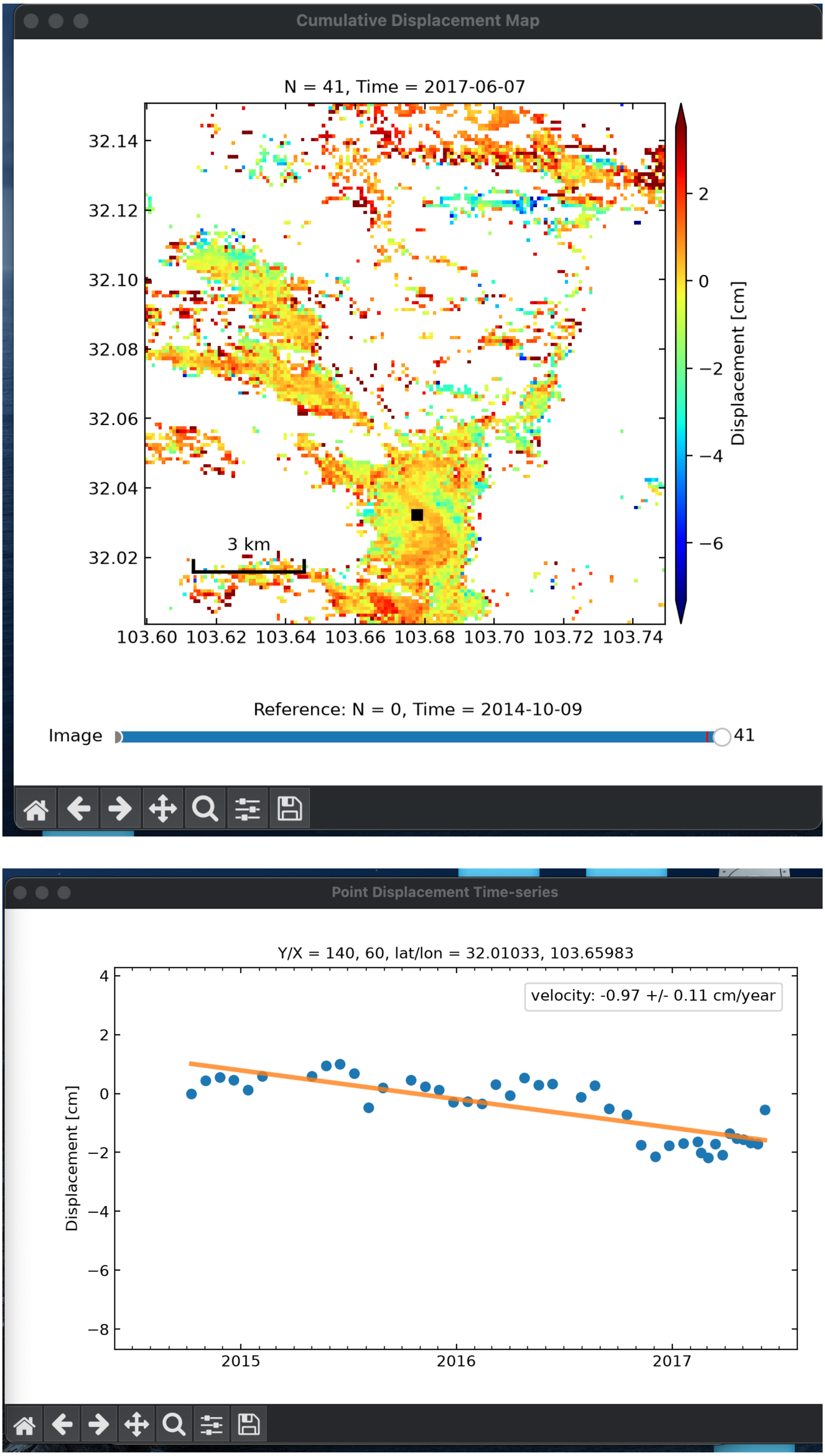

Figure 1: Cumulative Displacement Map (top) showing the spread of the movement, and Point Displacement Time-series (bottom) from the center of the landslide, showing the steady downward movement before the collapse.

The Cumulative Displacement map (top) shows a clear area of movement (warm colors) right where the slope eventually collapsed.

By looking at the time-series data for a single pixel in this active area (bottom), we can see how the movement changed over time. The graph shows a clear downward trend (orange line) starting in late 2014 and continuing right up to the June 2017 landslide.

During this time, the total movement reached about -2 cm to -6 cm compared to the stable reference point. This steady movement, captured by the SAR data, proves that the slope was slowly sliding long before the final, sudden collapse.